Simulation Report: Regime-Conditioned CVaR-QAOA on Nifty 50

Auto-generated by the notebook on 2026-10-11 | mode FULL

1. Setup and validation

Data: 18 NSE tickers, 2018-01-01 to 2026-10-09 (2167 trading days); frozen universe of N = 11 stocks, K = 5; 27 walk-forward refit points (every 63 days after 504 days of training), rebalance every 21 days, 10 bps one-way cost. Mode: FULL, data: snapshot.

• Leakage test passed: filtered HMM posteriors do not change when the future changes; smoothed posteriors do (so they are never used).

• Ising energies equal the QUBO cost on random bitstrings; penalty version equals cost + P(sum x - K)^2; QAOA optimum equals brute force.

• Qiskit circuits (Dicke state, XY-QAOA, penalty-QAOA, VQC) agree with the fast numpy simulator to numerical precision.

• SA never beats brute force; probabilities sum to 1; regime variances are ordered.

2. Headline conclusions (generated from this run's numbers)

• Constraint handling (RQ2): Dicke+XY samples were feasible 100% of the time at p=3, versus 35% for penalty+X mixer (CVaR) and 37% (expectation).

• Solution quality (RQ2, p=3): expected approximation ratio 0.692 (XY, CVaR) vs 0.741 (XY, expectation) vs 0.756 (penalty, CVaR); probability of the exact optimum 0.028 vs 0.0022 for uniform feasible sampling (lift 13.1x).

• Against random feasible sampling with the same 64 shots, the best-of-shots approximation ratio differs by +0.058 (0.994 vs 0.936): the QAOA sampler is ahead at this budget, but with only C(N,K) = 462 candidate subsets a classical search enumerates every candidate exactly, so this is not a practical advantage.

• Warm starts (RQ3): regime-blended library angles start at a normalised optimality gap of 0.66 (random cold start 1.22; 1 = random feasible subset, 0 = exact optimum) and sit at 0.61 vs 0.78 after 25 objective evaluations (paired Wilcoxon p = <0.0001).

• Control (shuffled-regime library) vs correct regime-keyed library, gap@25: 0.59 vs 0.61 (paired diff -0.020, p = 0.0970): no statistically clear advantage of regime keying over this control: the benefit is mainly from having reasonable starting angles.

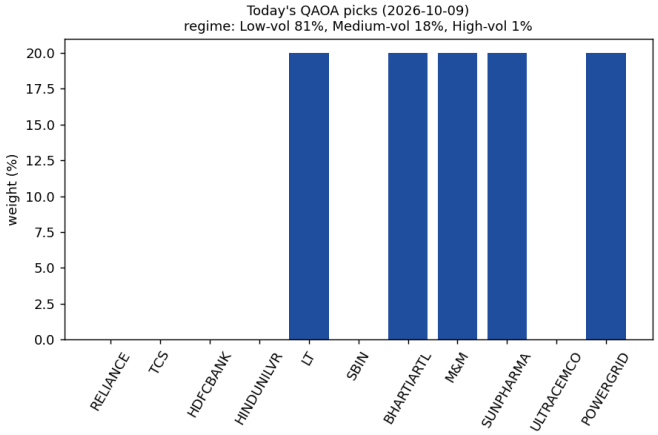
• Control (pooled library) vs correct regime-keyed library, gap@25: 0.63 vs 0.61 (paired diff +0.014, p = 0.0607): no statistically clear advantage of regime keying over this control: the benefit is mainly from having reasonable starting angles.

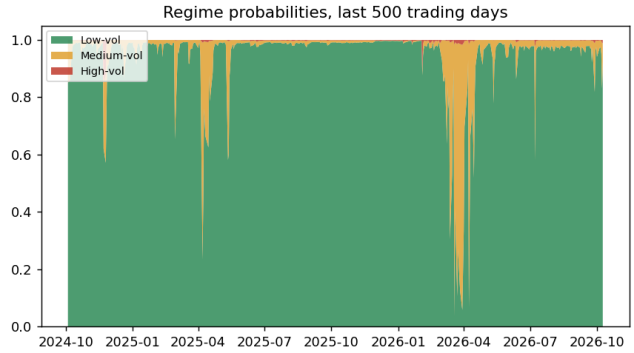
• Regime models (RQ1): HMM (classical): agreement with HMM 100%, switch rate 0.026, Spearman(E[regime], forward vol) 0.102; VQC (quantum): agreement with HMM 99%, switch rate 0.017, Spearman(E[regime], forward vol) 0.274; Threshold rule (non-quantum): agreement with HMM 42%, switch rate 0.070, Spearman(E[regime], forward vol) 0.509; VQC (quantum, realised-vol target): agreement with HMM 44%, switch rate 0.196, Spearman(E[regime], forward vol) 0.361.

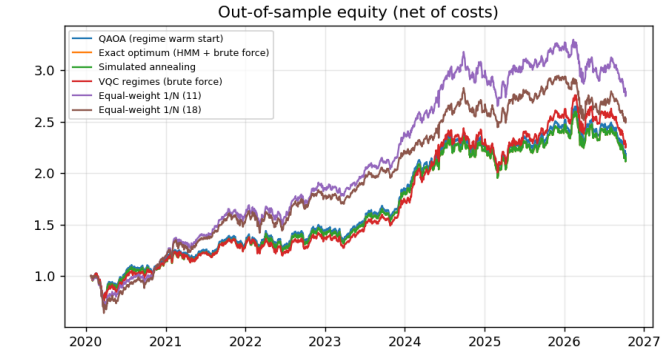
• Backtest: out-of-sample Sharpe of the exact optimum is 0.724 and of QAOA (regime warm start) 0.74 vs 1.003 for equal weight over the same 11 stocks. Strategies whose Sharpe differs from equal weight at the 5% level (block bootstrap): ['Regimes: Threshold rule (non-quantum) (brute force)', 'Regimes: VQC (quantum, realised-vol target) (brute force)'].

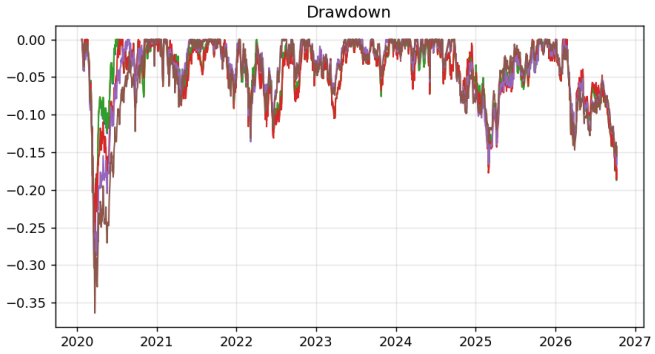
• Decision agreement with the exact optimum (same-set rate): QAOA cold (1 start): 89%, QAOA cold (3 starts): 100%, QAOA warm: dominant regime: 98%, QAOA warm: regime blend: 99%, QAOA warm: previous date: 100%.

• No quantum advantage is claimed: at N = 11 brute force solves the problem exactly in milliseconds; scaling results show classical search stays cheap while exact simulation of the quantum circuit doubles per qubit.

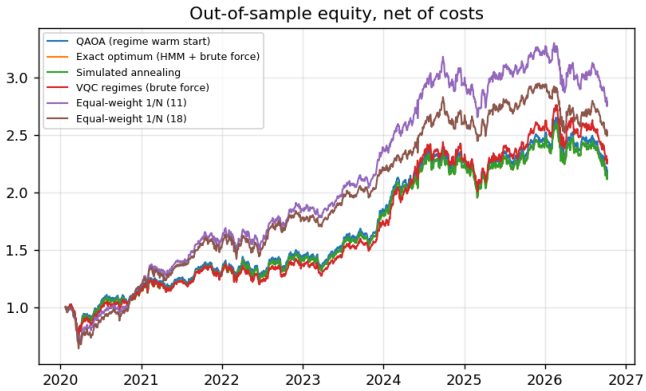

Today's QAOA picks (2026-10-09)
regime: Low-vol 81%, Medium-vol 18%, High-vol 1%
weight (%)
RELIANCE
TCS
HDFCBANK
HINDUNILVR
LT
SBIN
BHARTIARTL
M&M
SUNPHARMA
ULTRACEMCO
POWERGRID

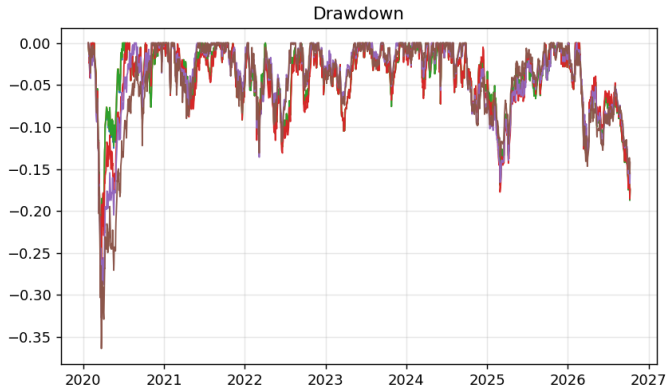

Regime probabilities, last 500 trading days
Low-vol
Medium-vol
High-vol


Out-of-sample equity (net of costs)
QAOA (regime warm start)
Exact optimum (HMM + brute force)
Simulated annealing
VQC regimes (brute force)
Equal-weight 1/N (11)
Equal-weight 1/N (18)


Drawdown

3. Backtest (Section 8)

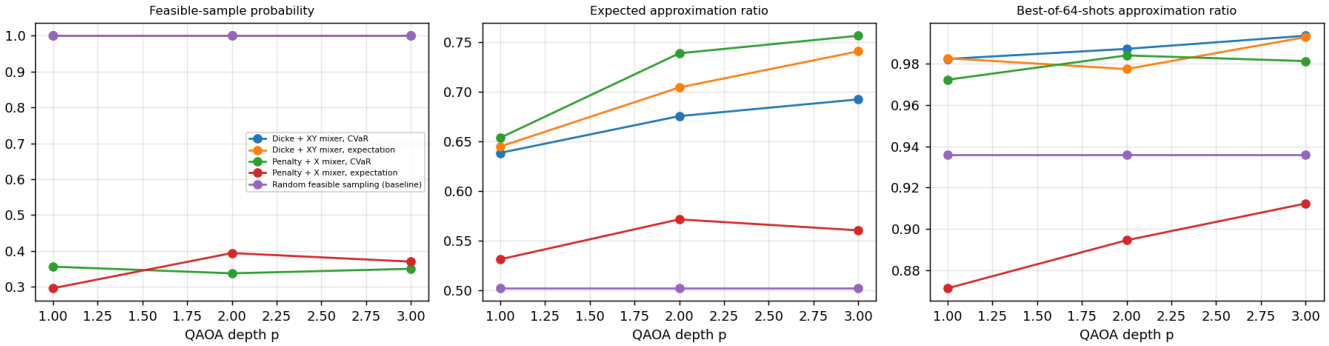




Strategy	Sharpe	AnnReturn	MaxDD	TrackErr	AvgTurnover	CI95	dSharpe_vs_EW11	dCI95	p_value
HMM + Brute force	0.724	0.1228	-0.29	0.0781	0.173	[0.04, 1.56]	-0.278	[-0.70, 0.08]	0.127
HMM + Simulated annealing	0.724	0.1228	-0.29	0.0781	0.173	[0.04, 1.56]	-0.278	[-0.70, 0.08]	0.127
QAOA cold (1 start)	0.698	0.1175	-0.29	0.078	0.193	[0.02, 1.54]	-0.305	[-0.75, 0.06]	0.103
QAOA cold (3 starts)	0.724	0.1228	-0.29	0.0781	0.173	[0.04, 1.56]	-0.278	[-0.70, 0.08]	0.127
QAOA warm: dominant regime	0.744	0.1274	-0.292	0.0778	0.158	[0.06, 1.57]	-0.259	[-0.69, 0.07]	0.133
QAOA warm: regime blend	0.74	0.126	-0.292	0.0784	0.168	[0.05, 1.57]	-0.262	[-0.70, 0.08]	0.144
QAOA warm: previous date	0.724	0.1228	-0.29	0.0781	0.173	[0.04, 1.56]	-0.278	[-0.70, 0.08]	0.127
CONTROL warm: shuffled-regime library	0.724	0.1228	-0.29	0.0781	0.173	[0.04, 1.56]	-0.278	[-0.70, 0.08]	0.127
CONTROL warm: pooled (regime-agnostic) library	0.724	0.1228	-0.29	0.0781	0.173	[0.04, 1.56]	-0.278	[-0.70, 0.08]	0.127
Regimes: VQC (quantum) (brute force)	0.771	0.1333	-0.292	0.0777	0.168	[0.06, 1.66]	-0.232	[-0.64, 0.10]	0.172
Regimes: Threshold rule (non-quantum) (brute force)	0.679	0.1081	-0.284	0.0719	0.542	[-0.04, 1.55]	-0.323	[-0.68, -0.02]	0.039
Regimes: VQC (quantum, realised-vol target) (brute force)	0.649	0.1021	-0.277	0.0793	0.152	[-0.04, 1.47]	-0.354	[-0.81, -0.01]	0.044
Equal-weight 1/N (all 11)	1.003	0.1683	-0.327	0.0	0.0	[0.23, 1.98]	0.0	[0.00, 0.00]	1.0
Equal-weight 1/N (all 18 downloaded stocks)	0.885	0.1511	-0.364	0.0445	0.0	[0.09, 1.88]	-0.118	[-0.31, 0.06]	0.156

Strategy	same_set	mean_overlap
HMM + Simulated annealing	1.0	1.0
QAOA cold (1 start)	0.887	0.978
QAOA cold (3 starts)	1.0	1.0
QAOA warm: dominant regime	0.975	0.995
QAOA warm: regime blend	0.988	0.997
QAOA warm: previous date	1.0	1.0
CONTROL warm: shuffled-regime library	1.0	1.0
CONTROL warm: pooled (regime-agnostic) library	1.0	1.0
Regimes: VQC (quantum) (brute force)	0.688	0.922
Regimes: Threshold rule (non-quantum) (brute force)	0.1	0.593
Regimes: VQC (quantum, realised-vol target) (brute force)	0.05	0.72

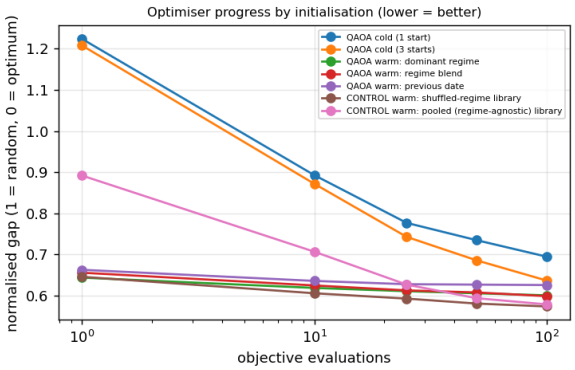
4. RQ2: XY + Dicke + CVaR vs penalty QAOA

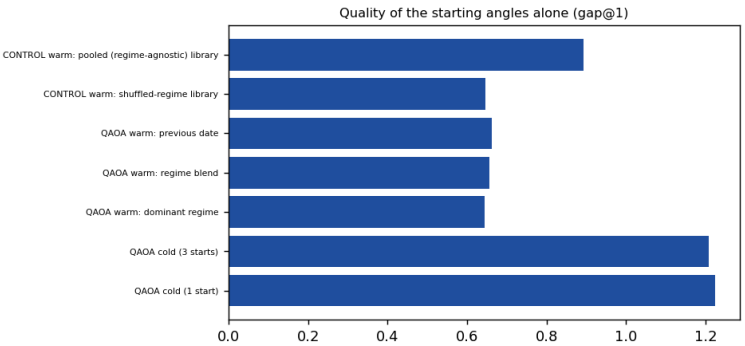


config	p	feasible_prob	p_opt	exp_AR	best_of_shots_AR	nfev	opt_lift_vs_random
Dicke + XY mixer, CVaR	1	1.0	0.011	0.638	0.982	191.0	5.1
Dicke + XY mixer, CVaR	2	1.0	0.024	0.675	0.987	378.708	11.1
Dicke + XY mixer, CVaR	3	1.0	0.028	0.692	0.994	393.708	12.9
Dicke + XY mixer, expectation	1	1.0	0.009	0.645	0.983	175.542	4.2
Dicke + XY mixer, expectation	2	1.0	0.015	0.704	0.978	359.875	6.9
Dicke + XY mixer, expectation	3	1.0	0.028	0.741	0.993	400.0	12.9
Penalty + X mixer, CVaR	1	0.356	0.014	0.653	0.972	164.833	6.5
Penalty + X mixer, CVaR	2	0.338	0.033	0.739	0.984	359.0	15.2
Penalty + X mixer, CVaR	3	0.35	0.03	0.756	0.981	398.875	13.9
Penalty + X mixer, expectation	1	0.296	0.002	0.531	0.871	117.708	0.9
Penalty + X mixer, expectation	2	0.394	0.008	0.571	0.894	370.083	3.7
Penalty + X mixer, expectation	3	0.37	0.007	0.56	0.912	399.833	3.2
Random feasible sampling (baseline)	1	1.0	0.002	0.502	0.936	0.0	0.9
Random feasible sampling (baseline)	2	1.0	0.002	0.502	0.936	0.0	0.9
Random feasible sampling (baseline)	3	1.0	0.002	0.502	0.936	0.0	0.9

config	p	depth	cx	total_gates
Penalty + X mixer	1	65	110	253
Dicke + XY mixer	1	781	472	1502
Penalty + X mixer	2	103	220	473
Dicke + XY mixer	2	866	604	1848
Penalty + X mixer	3	141	330	693
Dicke + XY mixer	3	951	736	2194

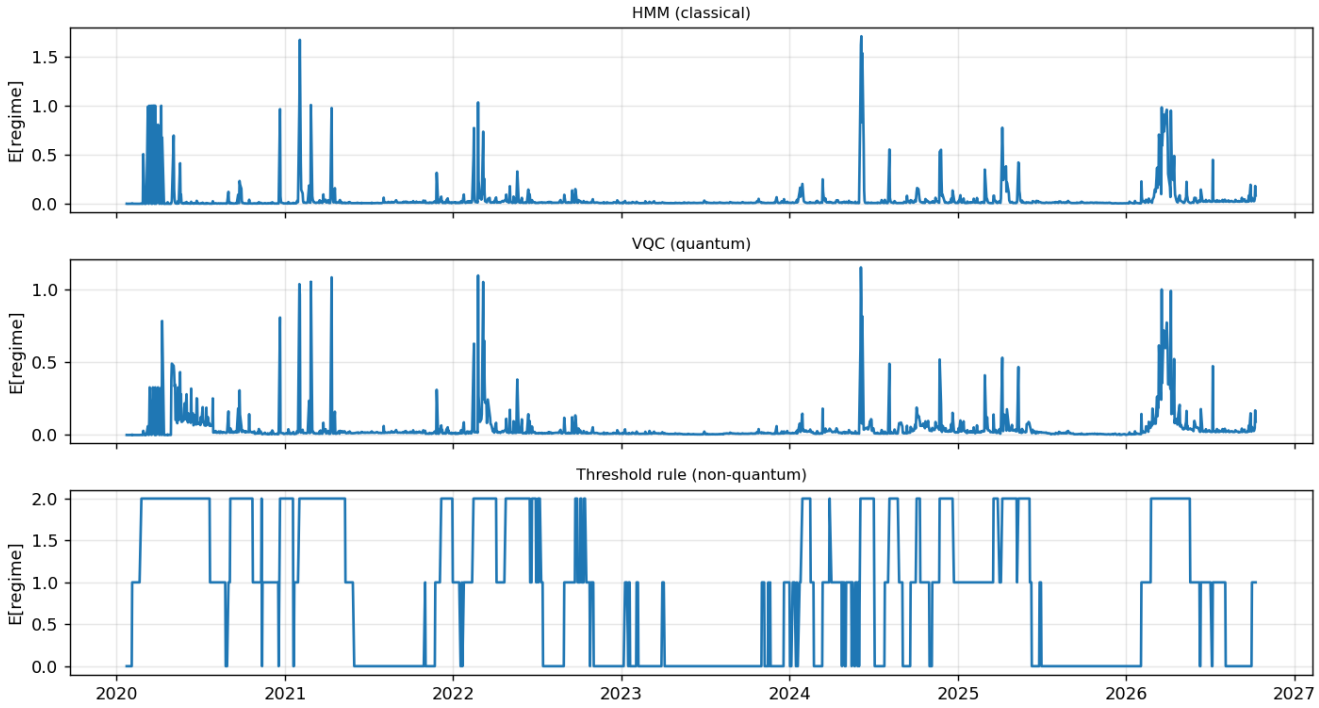
5. RQ3: regime-keyed warm starts (with controls)





Initialisation	gap@1 (start angles)	gap@10	gap@25	gap@50	gap@100	dates reaching 50% gap	p_opt	exp_AR	dgap@1 vs cold-1	Wilcoxon p (gap@1)	dgap@25 vs cold-1	Wilcoxon p (gap@25)
QAOA cold (1 start)	1.224	0.893	0.777	0.735	0.695	0.05	0.026	0.651	nan	nan	nan	nan
QAOA cold (3 starts)	1.208	0.872	0.743	0.686	0.637	0.062	0.0394	0.682	-0.016	0.5171	-0.034	0.0041
QAOA warm: dominant regime	0.644	0.619	0.611	0.606	0.601	0.088	0.0459	0.694	-0.58	0.0	-0.166	0.0
QAOA warm: regime blend	0.656	0.625	0.613	0.608	0.599	0.088	0.046	0.694	-0.568	0.0	-0.164	0.0
QAOA warm: previous date	0.663	0.636	0.628	0.627	0.626	0.012	0.0268	0.677	-0.561	0.0	-0.149	0.0
CONTROL warm: shuffled-regime library	0.646	0.606	0.593	0.581	0.574	0.138	0.0462	0.704	-0.578	0.0	-0.184	0.0
CONTROL warm: pooled (regime-agnostic) library	0.892	0.707	0.627	0.594	0.579	0.138	0.0465	0.703	-0.331	0.0	-0.15	0.0

6. RQ1: quantum (VQC) vs classical (HMM) regimes

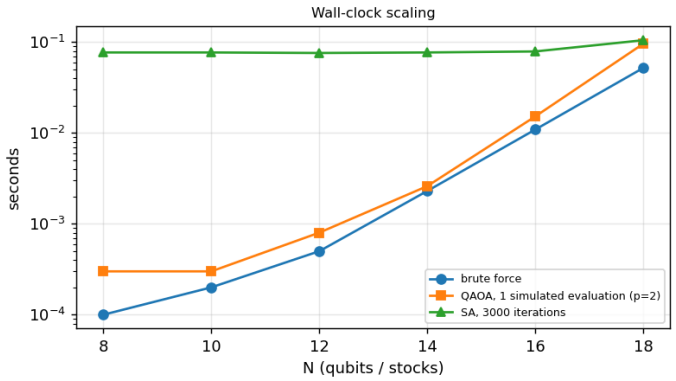


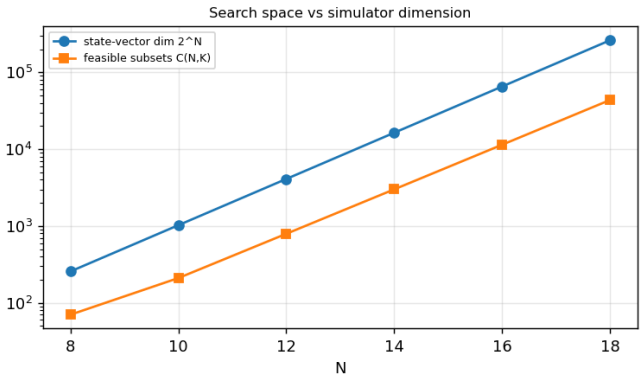
Regime source	Argmax agreement w/ HMM	Mean |dP| vs HMM	Switch rate (stability)	Spearman(E[regime], fwd vol)	Fwd vol | Low/Med/High
HMM (classical)	1.0	0.0	0.026	0.102	0.0086 / 0.0243 / 0.0078
VQC (quantum)	0.987	0.024	0.017	0.274	0.0089 / 0.0123 / nan
Threshold rule (non-quantum)	0.423	0.575	0.07	0.509	0.0066 / 0.0089 / 0.0121
VQC (quantum, realised-vol target)	0.435	0.63	0.196	0.361	0.0073 / 0.0093 / 0.0112

Strategy	Sharpe	AnnReturn	MaxDD	TrackErr	AvgTurnover
Regimes: HMM (classical) (brute force)	0.724	0.1228	-0.29	0.0781	0.173
Regimes: VQC (quantum) (brute force)	0.771	0.1333	-0.292	0.0777	0.168
Regimes: Threshold rule (non-quantum) (brute force)	0.679	0.1081	-0.284	0.0719	0.542
Regimes: VQC (quantum, realised-vol target) (brute force)	0.649	0.1021	-0.277	0.0793	0.152
Equal-weight 1/N (all 11)	1.003	0.1683	-0.327	0.0	0.0

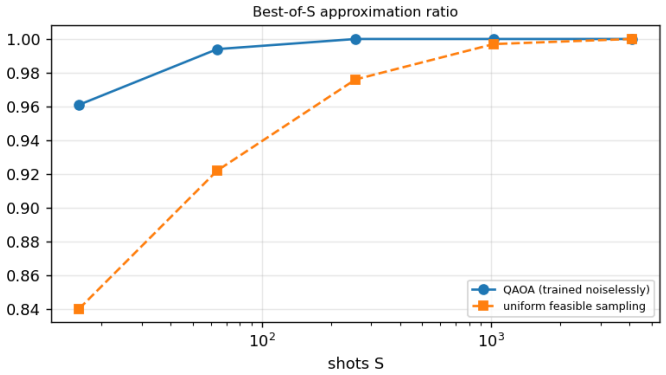
7. Robustness, scaling and sampling budget

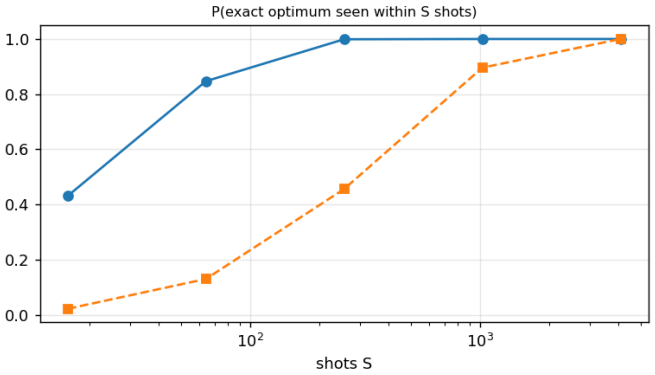
parameter	value	Sharpe	AnnReturn	MaxDD	TrackErr	AvgTurnover
K	3.0	0.583	0.1014	-0.282	0.1143	0.246
K	4.0	0.612	0.103	-0.303	0.0925	0.2
K	5.0	0.724	0.1228	-0.29	0.0781	0.173
K	6.0	0.813	0.1384	-0.27	0.0662	0.117
K	7.0	0.79	0.1328	-0.296	0.0534	0.098
lambda scale	0.5	0.716	0.1244	-0.292	0.0772	0.163
lambda scale	1.0	0.724	0.1228	-0.29	0.0781	0.173
lambda scale	2.0	0.677	0.1114	-0.29	0.0777	0.183
lambda scale	4.0	0.669	0.1078	-0.275	0.0774	0.183
kappa	0.0	0.691	0.1157	-0.29	0.0785	0.262
kappa	0.002	0.724	0.1228	-0.29	0.0781	0.173
kappa	0.005	0.698	0.1165	-0.29	0.0785	0.132
kappa	0.01	0.538	0.0849	-0.275	0.0794	0.098





N	K	feasible subsets C(N,K)	full state dim 2^N	state memory (MB)	brute force (s)	SA 3000 it (s)	SA hit optimum	QAOA build (s)	QAOA 1 eval p=2 (s)	Dicke prep CX
8	4	70	256	0.0	0.0001	0.077	True	0.013	0.0003	178
10	4	210	1024	0.02	0.0002	0.077	True	0.014	0.0003	246
12	5	792	4096	0.06	0.0005	0.076	True	0.019	0.0008	384
14	6	3003	16384	0.25	0.0023	0.077	True	0.027	0.0026	552
16	7	11440	65536	1.0	0.0109	0.079	True	0.079	0.0152	750
18	8	43758	262144	4.0	0.0519	0.105	True	0.355	0.0958	978





shots	qaoa_bestAR	random_bestAR	qaoa_P_opt	random_P_opt	P_opt lift (x)
16.0	0.961	0.84	0.431	0.022	19.59
64.0	0.994	0.922	0.847	0.13	6.52
256.0	1.0	0.976	0.999	0.456	2.19
1024.0	1.0	0.997	1.0	0.896	1.12

shots	qaoa_bestAR	random_bestAR	qaoa_P_opt	random_P_opt	P_opt lift (x)
4096.0	1.0	1.0	1.0	1.0	1.0

training_shots	exp_AR	p_opt
exact	0.6553	0.0212
1024	0.6105	0.0117
256	0.5238	0.0072
64	0.5203	0.0058

budget	SA mean AR	SA hit-optimum rate
100.0	0.999	0.925
500.0	1.0	1.0
3000.0	1.0	1.0

8. Today's decision

As of 2026-10-09: regime posterior Low-vol 81%, Medium-vol 18%, High-vol 1%.

Stock	QAOA pick	Exact (brute force)	Simulated annealing	Held at last rebalance	Exp. return (ann.)	Vol (ann.)
LT	1	1	1	1	0.181	0.267
SBIN	0	0	0	1	0.18	0.309
BHARTIARTL	1	1	1	1	0.189	0.288
M&M	1	1	1	1	0.222	0.315
SUNPHARMA	1	1	1	0	0.17	0.267
POWERGRID	1	1	1	1	0.177	0.259

9. Limitations

• Survivorship bias: today's constituents; universe of 11 stocks.

• HMM sees only the equal-weight market return; no Nifty index series in the bundled snapshot.

• VQC (HMM target) distils the HMM and cannot be claimed to beat it.

• Quantum results are ideal-state-vector simulations; no device noise. Defaults for K, lambda, kappa were fixed in advance, not tuned.

• Sharpe uses rf = 0; research prototype, not investment advice.